Methods for maintaining immobilization

External Devices

- Splint
- Brace
- Case
- External fixator
- Bandage

Internal devices

- Nails
- Plates
- Screws
- Wires
- Rods

Maintaining and Restoring function

- Maintain reduction and immobilization
- Elevate to minimize swelling
- Control anxiety and pain
- Isometric and muscle setting exercise

Complications of fractures

- Fat embolism
- Infection
- Hypovolemic or traumatic shock
- Delayed healing
- Nerve or blood vessel damage
- Arthritis
- Unequal leg length
- Mal union or non union

Nursing Management of fractures

- Bed rest
- Vital signs
- Skin care
- Prevent infection
- Intake and output chart
- Encouraging mobility
- Providing adequate nutrition.

Surgical Management

- Open reduction
- Closed reduction
- Internal fixation
- Bone graft
- Arthroplasty
- Joint replacement
- Amputation

2.5 Osteoporosis

Osteoporosis is a disease where increased bone weakness increases the risk of breakage of bone. Osteoporosis literally leads to abnormally porous bone that is compressible, like a sponge. It is the most common reason for a broken bone among the elderly. The most common site include the vertebrae in the spine, the bones of the forearm, and the hip.

Definition

Osteoporosis is a condition characterized by a decrease in the density of bone, decreasing its strength and resulting in fragile bones.

Causes

1. Genetic factor
2. Vitamin D and Deficiency factor
3. Post Menopausal women but may also occur in men
4. Chronic illness e.g. Malabsorption syndromes and Renal failures
5. Chemotherapy such as Cortico steroids
6. Cigarette smoking, alcohol consumption
7. Lack of exercise
8. Body type- small frame / short stature, low body fat.

Signs and Symptoms

1. Asymptomatic until later stages
2. Fracture after minor trauma may be first indications
3. Most frequent fractures associated with Osteoporosis include fractures of the distilled radius, spinal vertebrae, humorous pelvis

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